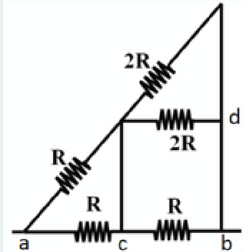
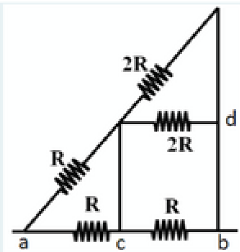
2



In this resistors circuit Vab= 160V and R= 20 Ohm. What is the equivalent resistor of the circuit ?
(4 Puan)

- 120/11 Ohm
- 40 Ohm
- 40/3 Ohm
- 20 Ohm
- 10 Ohm

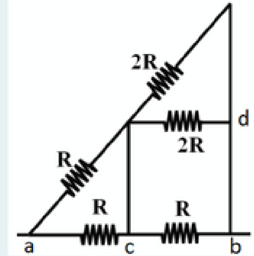
3



In this resistors circuit Vab= 160V and R= 20 Ohm. What is the total power of this circuit?
(4 Puan)

- 2349 W
- 2560 W
- 3200 W
- 1280 W
- 5120 W

4

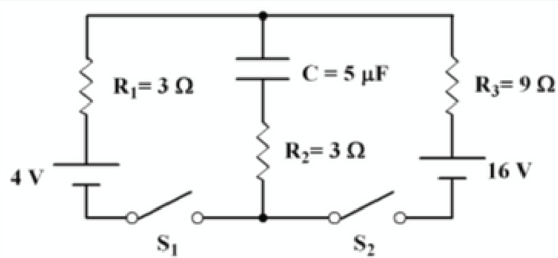


In this resistors circuit Vab= 160V and R= 20 Ohm. What is the current on R between cb points?
(4 Puan)

- 2 A
- 1.6 A
- 4 A
- 8 A
- 1 A

5

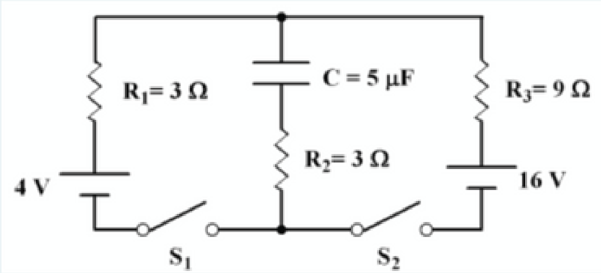
A circuit given in the figure is used for charging the 5 µF capacitor.
What is the absolute value of the current on the 4 V battery long after both of the switches (S1 and S2) are closed?
(4 Puan)



- 1 A
- 0.5 A
- 10/3 A
- 10/6 A
- 2 A

6

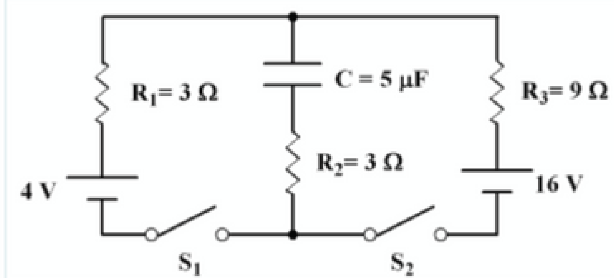
A circuit given in the figure is used for charging the 5 μF capacitor.
What is the charge on the capacitor long after the switches are closed?
(4 Puan)



- $25\mu C$
- $35\mu C$
- $40\mu C$
- $55\mu C$
- $90\mu C$

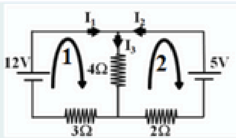
7

A circuit given in the figure is used for charging the 5 μF capacitor. S2 switch will be OPEN.
What is the time constant of the circuit if only S1 switch is closing?
(4 Puan)



- $7.5\ \mu s$
- $30\ \mu s$
- $6\ \mu s$
- $15\ \mu s$
- $75\ \mu s$

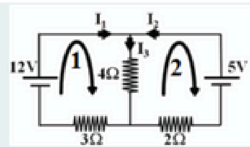
8



In the circuit given in the figure what is the equation from the 2nd loop?
(4 Puan)

- $-5 + 2I_2 + 4I_3 = 0$
- $5 + 2I_2 + 4I_3 = 0$
- $-5 - 2I_2 + 4I_3 = 0$
- $-5 + 2I_2 - 4I_3 = 0$
- $5 + 4I_2 + 2I_3 = 0$

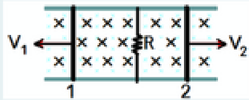
9



In the circuit given in the figure what are values of current I1 : I2 : I3
(4 Puan)

- 2A : -0.5A : 1.5 A
- 2A : 0.5A : 1.5 A
- 1.5A : 0.5A : 2 A
- -2A : 0.5A : -1.5 A
- 1.5A : 0.5A : 2 A

10

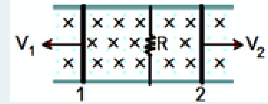


There are two parallel metal bars where a resistance R is connected to both of them. There are also two movable wires, the 1st one moves towards left with speed V1 and the 2nd one moves towards right with speed V2. Length of movable wires are L. There is also a constant magnetic field B into the page. There is NO resistance on the system except resistance R and there is NO frictional force in the system.
What are the magnetic flux change (manyetik aki degisimi) in the LEFT loop (between 1st wire and the resistance R.) and the in RIGHT loop (between the 2nd wire and the resistance R) respectively? LEFT : RIGHT
(4 Puan)

$R = 2\Omega : V_1 = 40m/s : V_2 = 60m/s : L = 50cm : B = 0.2T$

- $4\,V : 6\,V$
- $40\,V : 60\,V$
- $2\,V : 3V$
- $1\,V : 1.5\,V$
- $20\,V : 30\,V$

11

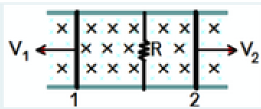


There are two parallel metal bars where a resistance R is connected to both of them. There are also two movable wires, the 1st one moves towards left with speed V1 and the 2nd one moves towards right with speed V2. Length of movable wires are L. There is also a constant magnetic field B into the page. There is NO resistance on the system except resistance R and there is NO frictional force in the system.
What is the direction of the INDUCED CURRENT on the 1st and 2nd wire respectively? 1st : 2nd
(4 Puan)

$R = 2\Omega : V_1 = 40m/s : V_2 = 60m/s : L = 50cm : B = 0.2T$

- *UP : DOWN*
- *UP : UP*
- *DOWN : UP*
- *DOWN : DOWN*
- *There is NO Current on each wire !*

12

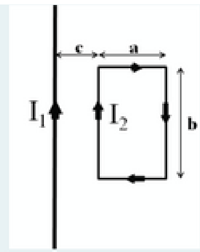


There are two parallel metal bars where a resistance R is connected to both of them. There are also two movable wires of lenght L, the 1st one moves towards left with speed V1 and the 2nd one moves towards right with speed V2. There is also a constant magnetic field B into the page. There is NO extra resistance on the system except resistance R and there is NO frictional force in the system.
What is the direction and the magnitude of the current on resistance R ?
(4 Puan)

$R = 2\Omega : V_1 = 40m/s : V_2 = 60m/s : L = 50cm : B = 0.2T$

- *UP : 1 A*
- *UP : 5 A*
- *DOWN : 1 A*
- *DOWN : 5 A*
- *No Current !*

13



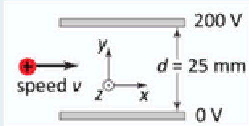
There is very long wire carrying a current of I1. c distance away there is a rectangular loop which carries a current of I2. The dimensions of rectangular loop is a and b.

i) find the direction and magnitude forces on each sides of the rectangular loop.

ii) find the total force acting on the loop. Is it zero?

iii) Imagine I2 =0 and rectangular loop is moving LEFT with a constant velocity v (getting far from long wire)
--) What is the direction of the INDUCED current on rectangular loop ? ---) What is the induced current value if the resistance of the rectangular loop is R? (Anonim olmayan soru)
(28 Puan)

14



A proton enters the region between the two plates shown in the figure with a speed v = 2000000 m/s
in the x-direction. The potential of the top and bottom plate is 200 V and 0 V, respectively, and their distance is d = 25 mm. What is the direction and magnitude of the magnetic field B required
between the plates for the proton to continue traveling in a straight line along the x-direction?
(4 Puan)

$V = 2 \times 10^6 \frac{m}{s}$

- Magnetic field cannot keep the proton traveling straight.
- $(2.5x10^{-6}\hat{j})T$
- $(-0.004\hat{k})T$
- $(-2.5x10^{-6}\hat{k})T$
- $(0.004\hat{j})T$

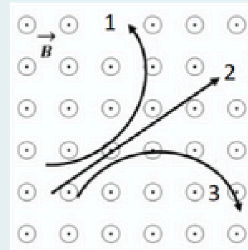
15

Two negatively charged particles, q1 and q2 , are of a distance r apart with q1 = 2q2. F21 is the force q2 exerts on q1 and F12 is the force q1 exerts on q2. What is the ratio of these two forces F12/F21 ?
(4 Puan)

$\frac{\vec{F}_{12}}{\vec{F}_{21}} = ?$

- $\frac{1}{2}$
- 2
- -1
- 1
- -2

16



3 different particle having the same velocity enter constant magnetic field and follow the paths given in the figure. B magnetic field direction is out of the page. What are the electric charge types of these particles in 1 : 2 : 3, respectively.
(4 Puan)

- Positive : Neutral : Negative
- Neutral : Positive : Negative
- Negative : Neutral : Positive
- More information is needed to answer the question !
- Negative: Positive: Neutral

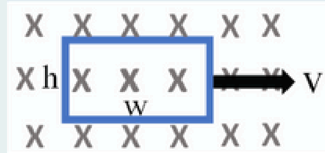
17

For an electron moving in a direction opposite to the electric field
(4 Puan)

$U = qV$

- its potential energy increases and its electric potential increases.
- its potential energy increases and its electric potential decreases.
- its potential energy decreases and its electric potential decreases.
- both the potential energy and the electric potential remain constant.
- its potential energy decreases and its electric potential increases.

18



A rectangular metal loop is pulled with a constant velocity V of 5m/s towards right in a constant magnetic field B of 0.5 Tesla which is into the page. The rectangular loop has width w=4cm and height h=2cm. What is the value for induced EMF and the direction of the induced current ?
(4 Puan)

CW: Clockwise(saat yönü)
CCW: CounterClockWise (saat yönü tersi)

- 0.4 mV ; CCW
- 0.4 mV ; CW
- 0 V ; No current !
- 2 mV : CW
- 2 mV : CCW

19

If a piece of wire is stretched by 2 times (its final length becomes 2 times of its initial length) what
happens to its resistance? (Density of the wire is constant)
(4 Puan)

- increases 2 times (twice)
- decreases 2 times (twice)
- stays constant
- increases 4 times
- decreases 4 times